

NANDESH MULGE

Software Engineer | Python Backend Developer • AI / GenAI Integration

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PROFESSIONAL SUMMARY

Software Engineer with professional experience in backend software engineering, relational database design, and secure RESTful API architectures using **Python**. Proven expertise in building normalized database schemas, optimizing complex SQL queries, and implementing defensive server-side logic across enterprise and retail domains. Specializing in modern asynchronous backend architectures with **FastAPI**, containerized workflows, and **AI/GenAI integration** (RAG pipelines, embeddings, pgvector, and LLM tool calling).

TECHNICAL SKILLS

Languages:	Python, SQL, HTML5, CSS3
Backend:	RESTful API Design, Pydantic, SQLAlchemy ORM, JWT Authentication, RBAC, CRUD Development
Frameworks:	FastAPI, Flask
Database:	PostgreSQL, MySQL, pgvector, Relational Schema Design (ERD), Query Optimization
AI / GenAI:	LLM API Integration, Embeddings, RAG Architectures, Vector Search, Tool / Function Calling
Testing:	pytest, Unit Testing, API Integration Testing, Mocking, Test Fixtures
Docker / Engineering:	Docker, Docker Compose, Containerization, Git, GitHub, Linux Environments

PROFESSIONAL EXPERIENCE

Infosys — System Engineer Bengaluru, India | Nov 2025 – May 2026

- Backend Architecture:** Developed and maintained core backend software architectures using Python and MySQL for complex enterprise-level business applications.
- Database Optimization:** Engineered structured SQL queries, relational joins, and database indexing strategies, significantly reducing query execution latency across automated reporting pipelines.
- Application Reliability:** Authored defensive programming modules, comprehensive server-side input validation configurations, and automated debugging routines to enhance system stability and reliability.
- SDLC & Code Quality:** Participated in structured Agile iterations, peer code reviews, and comprehensive technical documentation following corporate clean-code methodologies.

Ace Turtle — Python Developer Intern Bengaluru, India | Jan 2025 – May 2025

- REST API Development:** Constructed, tested, and deployed secure Flask REST API endpoints to streamline high-throughput data exchange between client applications and relational databases.
- Data Pipelines:** Co-engineered automated backend data transformation and ETL extraction pipelines, enhancing the integrity, processing speed, and reporting accuracy for retail analytics.
- Relational Schema Design:** Designed and normalized relational database schemas to efficiently store and scale high-volume transactional logs, retail store databases, and activity layers.

TECHNICAL PROJECTS

Job Assistant — AI-Powered Job Search & Interview Assistant *Python, FastAPI, PostgreSQL, SQLAlchemy, pgvector, Docker, LLM APIs* **IN PROGRESS**

- Backend & Auth Architecture:** Architecting a modular FastAPI backend with SQLAlchemy ORM, Pydantic schemas, and JWT authentication to manage candidate profiles, parsed resumes, and job application tracking.
- RAG & Vector Retrieval:** Implementing a grounded RAG pipeline using PostgreSQL and pgvector to chunk, embed, and semantically match resume experience profiles against specific job descriptions.
- LLM Tool Calling & Assessment:** Integrating LLM function calling to dynamically invoke backend services (skill gap calculations, status tracking) and generate targeted mock interview questions with structured evaluation.

CineReserve — Scalable Transaction & Scheduling Engine *Python, Flask, MySQL, HTML5, CSS3*

- Transactional Architecture:** Built an end-to-end seat reservation platform in Flask following MVC architecture, handling automated schedule rendering and real-time backend request validation.
- Access Control & Security:** Implemented Role-Based Access Control (RBAC) administrative dashboards and protected system endpoints through cryptographic session states and secure server-side verification policies.
- Database Integrity:** Designed normalized relational MySQL schemas with strict foreign-key mappings to preserve multi-table transaction states under concurrent request cycles.

EDUCATION

Sahyadri College of Engineering and Management Mangaluru, India | Graduated: 2025
Bachelor of Engineering (B.E.) in Computer Science and Engineering **CGPA: 8.20 / 10.00**